ADNOC GAS
LONG TERM - EXECUTE STAGE
PROJECT 5.0 (P5)
EPCm FOR HABSHAN DBN
COMPANY PROJECT No. 30201-50150-LT-HE
WORK ORDER NO. 3000076501

Comments Resolution Sheet (CRS) - EPCm

COMPANY Trans. No.:	**ADNOCGP-WTRAN-021503**	**Received Date:**	**23-Jan-26**
COMPANY Doc. No.:	**30201_50150_LT_HE-H4-280-17-23-003**	**Revision:**	**B**
Document Title:	**FOUNDATION GA & DETAILS SUBSTATION SS-203**	**Status Code:**	**2**
ADNOC PM:	**Dr Azharuddin Mohammed**	**EPCm. CONTRACTOR PM:**	**Shailesh**
ADNOC Reviewer:		**EPCm. CONTRACTOR Originator:**	**Ankita**

Review Codes: ☐ **Code-1: Rejected** ☑ **Code 2: Comments as Noted**
☐ **Code-3: No Comments** ☐ **Code 4: For Information Only**

REVISION A

No.	COMMENT SECTION/ CLAUSE / DRAWING REFERENCE	COMPANY Comment	COMPANY Reviewer	EPCm CONTRACTOR Response	Open/Closed	COMPANY Response (If Comments are still open / Not incorporated or if Eng Contractor Response Not Acceptable)	EPCm CONTRACTOR Response	Open/Closed	Remarks
1		HOLDS to be resolve before next submission.	Mr Rahul Shedage	Hold list shall be removed after approved Architectural drawings and other discipline drawings.					
2		Align with Materials and corrosion for the criticality. Also, align with CP vendor for the required details and execute during the foundation casting. Ensure to align the timeline with CP vendor.	Mr Vinodhkumar Ranganathan	Noted & incorporated		Ensure that CP vendor installation drawings available with GCC1 before the rebar installation on Blinding concrete. Also, ensure that CP vendor shall be available for training/monitoring the CP installation.	Noted		
3		EPCM to ensure this in Early works Earthworks scope itself for this area. check and update as required.	Mr Vinodhkumar Ranganathan	Noted & note has been removed	CLOSED	Noted			
4		Update to add the GWT in terms of level.	Mr Vinodhkumar Ranganathan	Noted & incorporated	OPEN	Open. Could not verify the compliance	GWT level is mentioned in calculation, based on that concrete protection type is mentioned in drawing.		
5		Ensure that the location of the building is approved by CPY HSE	Mr Vinodhkumar Ranganathan	CPY HSE to approve building location. Necessary update shall be performed if any modification required due to building location.	OPEN		CPY HSE YET TO APPROVE BUILDING LOCATION		
6	30201_50150_LT_HE-H4-280-17-23-003-02	Shall be confirmed based on the displacement and MAF of sealant.	Mr Vinodhkumar Ranganathan	Necessary calculation have been provided in calculation report		No, Attach the MAR	Construction contractor confirmed that material proposed will be available. SEE ATTACHED MAIL SNAPSHOT. **From:** 史海鹏 Shi Haipeng <shihp.snei@sinopec.com> **Sent:** 10 January 2026 14:32 **To:** Sanpui, Debjit <debjit.sanpui@woodplc.com> **Cc:** Rinaldi, Giovanni <giovanni.rinaldi@woodplc.com>; 王金鑫 Fred Wang <wangjinxin.snei@sinopec.com>; Naraf <Shailesh.Kshirsagar@woodplc.com>; 吴少华-南京工程 <wushaohua.snei@sinopec.com>; 周新文-南京工程 <zh **Subject:** RE: P5.0 HBASHAN DBN: Building clarification on materials Dear Debjit, The materials you proposed can be proceeded. Thank you. Best regards, Shi **From:** Sanpui, Debjit <debjit.sanpui@woodplc.com> **Sent:** Friday, January 9, 2026 10:09 PM **To:** 史海鹏 Shi Haipeng <shihp.snei@sinopec.com> **Cc:** Rinaldi, Giovanni <giovanni.rinaldi@woodplc.com>; 王金鑫 Fred Wang <wangjinxin.snei@sinopec.com>; Naraf <Shailesh.Kshirsagar@woodplc.com> **Subject:** P5.0 HBASHAN DBN: Building clarification on materials 提醒：这是一封外部邮件。请谨慎对待邮件内容、链接和附件，谨防钓鱼。 Reminder: This is an external email. Please treat email content, links, and attachments with caution and be caut Dear Shi Good evening! We need confirmation on below material & other details used for the project. 1. Technical Data Sheet – For sealant used for expansion joint 2. Confirmation on 32mm dia bars can be used 3. False ceiling max hanger supports Based on your confirmation, we shall use the subjects in our drawings and calculations. *T & WR* Dr. Debjit Sanpui, C. Eng. FHEI Project Manager, Wood Plc. Abu Dhabi Mobile no. +971 50 837 4695		
7		Through calculation it shall be established that sealant at expansion joint can accommodate the relative lateral movement. In some cases, shear wall may be required to meet the criteria. Typical comment in all the expansion joints.	Mr Vinodhkumar Ranganathan	Necessary calculation have been provided in calculation report			Refer above response		
8		For non-blast, width provided is high. optimize	Mr Vinodhkumar Ranganathan	Building width has been increased in revised layout, hence base pressure is critical. Optimizing width of foundation not feasible.		Noted. But still 3.0 is more for 13m wide building. Even with 2.6 to 2.7m gives good amount of reduction in concrete quantity. Check and update	Strip footing width is optimised to 2.7 m		
9		Update to show the chamfer at the reentrant corners. Typical comment	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				
10		why CF13? to avoid confusion, CF1? check and update	Mr Kiranbhai Kuntmal	Noted & incorporated	CLOSED				
11		As per Type 2a, 30mm thick screed and membrane is required. Update to mention the same. Typical comment	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				
12		Update to show the overlapping detail	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				
13		Update to show the expansion joint detail to execute	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				
14		check the correctness and update as required	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				

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15	30201_50150_LT_HE-H4-280-17-23-003-03	Shown levels are not okay. All grade beam levels to be reverified and update. Recommend to discuss with CPY in advance with the proposal.	Mr Vinodhkumar Ranganathan	Noted & incorporated	CLOSED				
16		The details shown are not correct. Update as required for transformer bay.	Mr Vinodhkumar Ranganathan	Noted & incorporated	OPEN	Previous comment not complied.	Detail Updated		
17		column to extend upto 101.850 ? check and update	Mr Vinodhkumar Ranganathan	Noted & revised as per the changes in the architectural drawings	CLOSED				
18		May not be required. check and update	Mr Vinodhkumar Ranganathan	Noted & revised	CLOSED				
19		check and limit to thickness < 1m.	Mr Vinodhkumar Ranganathan	Noted & revised	CLOSED				
20		T12 may be sufficient. Justify the basis.	Mr Vinodhkumar Ranganathan	Noted & revised	OPEN	Previous comment not complied.	Updated		
21	30201_50150_LT_HE-H4-280-17-23-003-04	The details shown are not correct. Update as required for transformer bay.	Mr Vinodhkumar Ranganathan	Noted & incorporated	OPEN	Previous comment not complied.	Detail updated		
22		Update to provide the top U-bar	Mr. Mohamed Elsadani	Noted & incorporated	closed				
23		Horizontal rebar shown is very heavy. check and update	Mr Vinodhkumar Ranganathan	Horizontal Rebar optimized	closed				
24		T16 @ 125	Mr Mahesh More	Noted & revised	closed				
25		Update to show the splice location. Also, check for maximum height allowed from foundation top without any support	Mr Vinodhkumar Ranganathan	Noted & revised	open	Previous comment not complied.	Tentative starter bar position shown,To be confirmed from contractor.		
26	30201_50150_LT_HE-H4-280-17-23-003-05	32 dia. rebar may not be available and difficult for bar bending. Avoid using 32 dia.rebar. Typical comment.	Mr Vinodhkumar Ranganathan	32 mm bar can be used as per mail confirmation from GCC1/GCC2 Contractor received dated 14-01-2026	closed				
27		Avoid placing stirrups adjacent to each other like shown. Concrete flow may be difficult. check and update. Typical comment.	Mr Vinodhkumar Ranganathan	Noted & revised	closed				
28		100 for T2 (typical comment)	Mr Vinodhkumar	Noted & incorporated	closed				
29		For platform columns, shown size seems to be high. check and update	Mr Vinodhkumar Ranganathan	Column size Optimized	closed				
30	30201_50150_LT_HE-H4-280-17-23-003-04	include detail	Mr Sreekanth Nuthanapati	Noted & Incorporated.	open	Previous comment not complied.	Detail included		
31	30201_50150_LT_HE-H4-280-17-23-003-02	Add a table without fail: Concrete Grade : Rebar Grade: Clear cover for foundation: Clear cover for Column: Clear cover for Grade beam: Clear cover for Fire wall:	Mr Vinodhkumar Ranganathan	Noted & incorporated	closed				
32	30201_50150_LT_HE-H4-280-17-23-003-02	1. Architectural drawings shall be in code 3 before the IFC of the foundation drawings. 2. Any update required based on the UG detection survey and slit trenching shall be carried out. 3. EPCM to reverify with the execute stage Geotechnical report before the Final submission. 4. Any update required based on the 3D model review comments shall be carried out. 5. Any update required based on the comments in structural and Foundation calculation report shall be carried out. 6. Ensure 3D model and drawings align each other.	Mr Vinodhkumar Ranganathan	1. Noted. 2. Necessary update in drawing shall be perfomed based on UG Detection survey and slit trenching. 3. Execute stage Geotech report is not available now. As per available Define stage Geotech report informations soil parameters has been considered, However necessary modification shall be done as per the execute stage Geotech report. 4. Noted. Update required as per 3D model review comments shall be incorporated. 5. Noted. 6. Noted		1. open 2. Noted 3. open. Expedite for the execute stage details and verify & confirm the foundation sizing before the construction starts. 4. open 5. Open. To be retained till the final report submission 6. Noted	1.Noted. 3.Execute stage report awaiting. 4. Noted. Update required as per 3D model review comments shall be incorporated. 5.Noted.		
REV B									
1	[1] 05	Update to show nos of bars at each face. Typ. comment.	Mr Mandar Nadgouda	Noted & Incorporated					
2	[1] 02	Update to correct Archi Drawing number.	Mr Mandar Nadgouda	noted & incorporated					
3	[1] 03	Check and update for Side bars for all footings.	Mr Mandar Nadgouda	Noted & incorporated					
4	[1] 03	Check thickness w.r.t Ldc of column rebars. Update as necessary	Mr Mandar Nadgouda	Checked and thickness updated.					
5	[1] 03	reinforcement position is not clear from the table. Update to show rebars in plan or appropriate section at suitable locations. Update as necessary.	Mr Mandar Nadgouda	Noted & Incorporated					
6	[1] 02	Size for clean agent room seems to be high. Check and update to optimize	Mr Vinodhkumar Ranganathan	foundation size optimised					
7	[1] 02	Typical comment : Align with E&I for the columns located in the E&I trench.	Mr Vinodhkumar Ranganathan	Discussed with E&I, trench layout updated					

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8	[1] 02	Attach the reference that the Cable trench shown is approved by respective discipline. Typical comment	Mr Vinodhkumar Ranganathan	Cable trench layout is based on electrical Input. Mail reference attached. RE: SS 203 Trench Layout/Transformer foundation Summarize Kumar, Sirish To: Chatterjee, Ankita Cc: Mahadik, Vishal; Chatterjee, Anirban; Beg, Rizwan; Harikrishnan, Balaji; Vadhry, Viswanath; Sharma, Anurag; Elsayed, Ibrahim; Hedalkar, Onkar; +2 others Reply Reply all Forward Tue 2026-01-27 18:44 You forwarded this message on Tue 2026-01-27 21:31 View conversation Dear Ankita, Please follow the below path for the ICR Cable trench routing layouts submitted for COMPANY review. O:\AE\AUH02\AUH02-Projects\ADNOCGAS\608200\GEN\03ENG\3.3_Elec\12. LAY\4. CRL\3. SS-203\A\ISSUED The same will be updated based on COMPANY comments and changes to the equipment arrangement. Expected readiness – 10th Feb, 26. Regarding the latest Equipment layout developed on IFA – Architecture drawing, we will be taking up F2F discussion with COMPANY starting tomorrow for the updated Equipment layout which will be issued as IFD with HOLDS as per the alignment with COMPANY. Path for the Rev-C Equipment layout working file : O:\AE\AUH02\AUH02-Projects\ADNOCGAS\608200\GEN\03ENG\3.3_Elec\12. LAY\1. EQP\3. SS-203\C\Working Dimensions for the Transformer Bay as indicated in the above Rev C drawing are final and will not undergo change. Details of Distribution TF are still awaited. Power TF details are final. Thanks Sirish Kumar					
9	[1] 02	check to avoid the trench inside the HVAC room. Discuss before the next submission	Mr Vinodhkumar Ranganathan	communicated to E&I, awaiting their response.					
10	[1] 03	Beam is missing. check and update	Mr Vinodhkumar Ranganathan	Noted & Incorporated					
11	[1] 03	For battery room and HVAC room, Grade beam Top may be required to match with FFL. Check and update	Mr Vinodhkumar Ranganathan	To be finalised yet, Hence Grade beam top on Hold					
12	[1] 03	F13 ??	Mr Vinodhkumar Ranganathan	Detail for F1 to F11 only.F12 and F 13 details are shown in next page.					
13	[1] 04	Chck the correctness and update	Mr Vinodhkumar Ranganathan	Updated					
14	[1] 04	Groove required. check and update. Typical comment	Mr Vinodhkumar Ranganathan	Groove updated in drawing					
15	[1] 04	Insert plates may be required to support the pipe supports, cable trays, bus ducts etc. check the requirements and update as required	Mr Vinodhkumar Ranganathan	Insert plate details will be incorporated upon receival of final information.					
16	[1] 04	Confirm that the width and level shown is approved by Piping. Attach the reference mail in the CRS.	Mr Vinodhkumar Ranganathan	Piping Transformar wall levels Currently Wall height under hold.					
17	[1] 04	3-T20 B/F ??	Mr Vinodhkumar Ranganathan	Noted & Incorporated					
18	[1] 05	along the thickness also required at a spacing equal to side face rebars	Mr Vinodhkumar Ranganathan	updated					
19	[1] 04	update to show the staggering and locations	Mr Vinodhkumar Ranganathan	updated					
20	[1] 04	Position of bars shown is not correct	Mr Vinodhkumar Ranganathan	updated					
21	[1] 05	Alternate bars shall be tied. check and update	Mr Vinodhkumar Ranganathan	Updated					
22	[1] 05	Why at one location this sizing is required? clarify	Mr Vinodhkumar Ranganathan	To provide clearance to electricalroom, column size is updated.					
23	[1] 03	Only at change in direction, this heavy may be required. check and update to optimize.	Mr Vinodhkumar Ranganathan	optimized					
24	[1] 03	Heavy rebar details required only at certain locations. check to optimize. Provide plan rebar layout with splicing locations.	Mr Vinodhkumar Ranganathan	optimized					
25	[1] 02	Update to show the joint detail or provide clear reference to where detail available to execute during the concrete casting.	Mr Vinodhkumar Ranganathan	updated					
26	[1] 03	Please refer the E08 type standard earthing installation from drawing 30201_50150_LT_HE-GG-000-16-21-001 for REINFORCEMENT CONCRETE STRUCTURE	Mr Kiranbhai Kuntmal	Noted ,drawing reference added					
27	[1] 03	Ensure all frame beams are shown appropriately in the section	Mr Vinodhkumar Ranganathan	noted and incorporated					
28	[1] 03	Side face rebars required with a spacing <=300mm. Update. Typical comment.	Mr Vinodhkumar Ranganathan	noted and incorporated					

DRAWING NO.	DRAWING TITLE	SHEET	SHEET TITLE
30201_50150_LT_HE-H4-280-17-23-003	UNIT 280 -HABSHAN 4 FOUNDATION GA & DETAILS SUBSTATION SS-203	01	DRAWING INDEX
		02	BUILDING FOUNDATION LAYOUT
		03	FOUNDATION SECTIONS & DETAILS
		04	FIRE WALL DETAILS
		05	COLUMN & CF1 DETAILS

SCALE 1:10
0 0.2 0.4 0.6 0.8 1.0m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRO
0	29-JAN-2026	ISSUED FOR CONSTRUCTION (IFC)	VA	KCH/BH	VV/MS
B	17-JAN-2026	ISSUED FOR APPROVAL (IFA)	JAM	KCH/BH	VV/MS
A	22-DEC-2025	ISSUED FOR COMPANY REVIEW (ICR)	SUT	KCH/BH	VV/MS

SCALE:	DRAWN:	CHKD:	APPRO:
1:10	SAULE	ANIRBAN/BALAJI	VISWANATH/SOFI

أدنوك للغاز
ADNOC Gas
ADNOC

EPCm CONTRACTOR: wood.
EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE
PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

DRAWING TITLE
UNIT 280 -HABSHAN 4
FOUNDATION GA & DETAILS SUBSTATION SS-203
DRAWING INDEX

UNIT: 262 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXX
ADNOC GAS CONTRACT No. 3000076501

ADNOC GAS PROJ. No.	ADNOC GAS DWG. No.	SHEET	REV
30201_50150_LT_HE	30201_50150_LT_HE-H4-280-17-23-003	01 OF 05	0

SHEET SIZE : A0

CAD NO.: 30201_50150_LT_HE-H4-280-17-23-003_C.dwg

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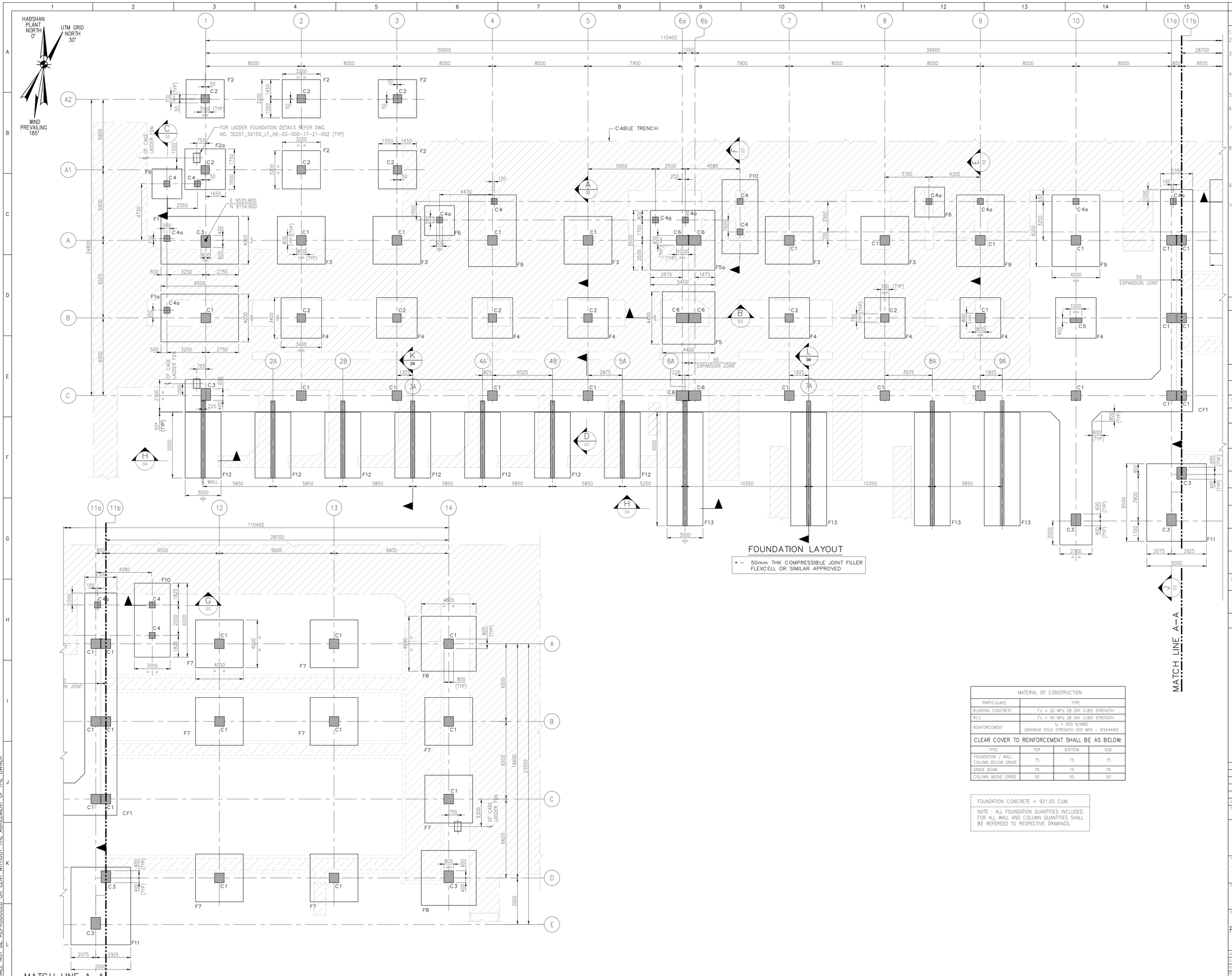


NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. THE COORDINATES MENTIONED ARE IN PLANT COORDINATES AND ELEVATIONS MENTIONED ARE IN JEBEL DHANNA CHART DATUM (JDCD)
4. FGL.EL.100.000 CORRESPONDS TO 73.000 ELEVATION BASED ON JEBEL DHANNA CHART DATUM (JDCD) AND 74.010 ELEVATION BASED ON NEW ABU DHABI DATUM (NADD).
5. PLANT COORDINATES E 9431.910, N 7903.650 EQUAL TO UTM COORDINATES E 769200.960, N 2637528.035.
6. FOR GENERAL NOTES, ABBREVIATIONS AND SYMBOLS – CONCRETE WORKS REFER STANDARD DRAWING NO. 30201_50150_LT_HE-GG-000-17-11-001.
7. FOR DETAILS OF PROTECTION TO CONCRETE FOUNDATIONS AND PEDESTALS REFER STANDARD DRAWING 30201_50150_LT_HE-GG-000-17-31-002.
8. THIS FOUNDATION IS CONSIDERED TO BE CRITICAL FOUNDATION INLINE SOW-30201_50150_LT_HE-GG-000-63-22-001 AND CORROSION MONITORING SYSTEM SHALL BE PROVIDED AS PER AGES STANDARD DRAWING AGES-STD-01-108. REBAR ELECTRICAL CONTINUITY SHALL BE CHECKED AND ENSURED BEFORE CONCRETE POURING.
9. EPCM CONSTRUCTION CONTRACTOR SHALL ENSURE THAT EXCAVATION & EXECUTION SHALL BE CARRIED OUT DURING NORMAL WIND CONDITIONS ONLY.
10. EXISTING UNDER GROUND SERVICES SHALL BE VERIFIED WITH SLIT TRENCH AND SHALL BE SUITABLY REROUTED IN AGREEMENT WITH COMPANY END-USER, BEFORE START OF WORKS.

FOUNDATION DESIGN ARE CARRIED OUT WITH 220KN/M² NET SBC AT 2.5M DEPTH AS PER DEFINE STAGE SOIL REPORT.

LEGEND

℄	CENTERLINE	FDN	FOUNDATION
TYP	TYPICAL	COL/C1	COLUMN
HPP	HIGH POINT OF PAVING	BOF	BOTTOM OF FOUNDATION
FGL	FINISHED GRADE LEVEL	TOC	TOP OF CONCRETE
THK	THICKNESS		
TOGR	TOP OF GRAVEL		

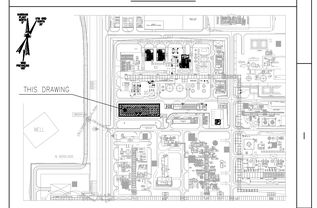
HOLD

1. CABLE TRENCH 2. BUILDING COORDINATE

REFERENCE DRAWINGS

30201_50150_LT_HE-H4-262-13-60-001	UNIT PLOT PLAN – UTILITY AREA HABSHAN-4 (U261, U262 ,U264, U266, U267, U268, U272, U280)
30201_50150_LT_HE-H4-280-20-22-002	UNIT 280 – HABSHAN 4 ARCHITECTURAL LAYOUT AND DETAILS SUBSTATION SS-203
30201_50150_LT_HE-GG-000-17-11-002	TYPICAL CONCRETE GA AND REINFORCEMENT DETAILS
30201_50150_LT_HE-GG-000-16-21-001	TYPICAL INSTALLATION DETAILS FOR EARTHING AND LIGHTING SYSTEM
30201_50150_LT_HE-H4-262-14-21-001	CIVIL GENERAL ARRANGEMENT AND UNDERGROUND COMPOSITE LAYOUT_HABSHAN 4 – UTILITY
30201_50150_LT_HE-H4-280-17-23-004	STRUCTURAL CONCRETE LAYOUT AND DETAILS FOR SUBSTATION SS-203
30201_50150_LT_HE-GG-000-17-31-002	DETAILS OF PROTECTION TO CONCRETE FOUNDATIONS AND PEDESTALS
DC-30201_50150_LT_HE-H4-280-17-22-002	STRUCTURAL CONCRETE – DESIGN SUBSTATION SS-203
30201_50150_LT_HE-GG-000-16-21-001	TYPICAL INSTALLATION DETAILS (EARTHING & LIGHTNING SYSTEM)

KEY PLAN



MATERIAL OF CONSTRUCTION

PARTICULARS	TYPE
BLINDING CONCRETE	f'c = 20 MPa 28 DAY CUBE STRENGTH
RCC	f'c = 40 MPa 28 DAY CUBE STRENGTH
REINFORCEMENT	fy = 500 N/MM2 (MINIMUM YEILD STRENGTH 500 MPA – BS44449)

CLEAR COVER TO REINFORCEMENT SHALL BE AS BELOW:

TYPE	TOP	BOTTOM	SIDE
FOUNDATION / WALL COLUMN BELOW GRADE	75	75	75
GRADE BEAM	75	75	75
COLUMN ABOVE GRADE	50	50	50

FOUNDATION CONCRETE = 921.00 CUM.

NOTE : ALL FOUNDATION QUANTITIES INCLUDED. FOR ALL WALL AND COLUMN QUANTITIES SHALL BE REFEREED TO RESPECTIVE DRAWINGS.

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	29-JAN-2026	ISSUED FOR CONSTRUCTION (IFC)	VA	KCh/BH	VV/MS
B	17-JAN-2026	ISSUED FOR APPROVAL (IFA)	JAM	KCh/BH	VV/MS
A	30-DEC-2025	ISSUED FOR COMPANY REVIEW (CR)	SUT	KCh/BH	VV/MS

SCALE: 1:100 | DRAWN: SAULE | CHKD: ANIRBAN/BALAJI | APPRD: VISWANATH/SOFI

ADNOC Gas

EPCm CONTRACTOR: wood. | EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5) LONG TERM - EXECUTE STAGE EPCm FOR HABSHAN DBN

DRAWING TITLE: HABSHAN 4 – UNIT 262 FOUNDATION GA & DETAILS SUBSTATION SS-203 BUILDING FOUNDATION LAYOUT

UNIT: 262 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No.: 30201_50150_LT_HE | ADNOC GAS DWG. No.: 30201_50150_LT_HE-H4-280-17-23-003 | SHEET: 02 OF 05 | REV: 0

SHEET SIZE : A0

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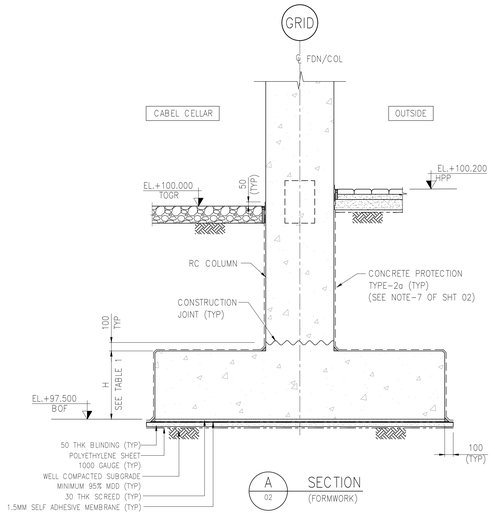
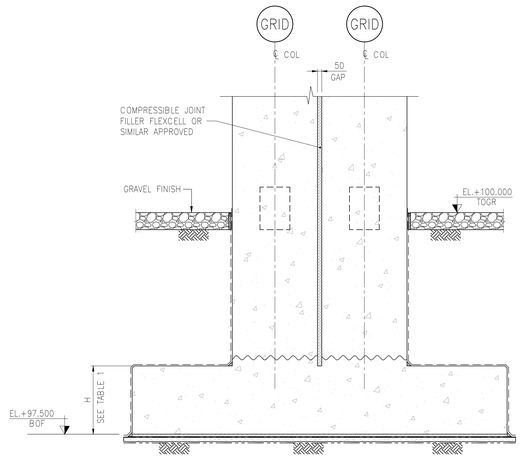
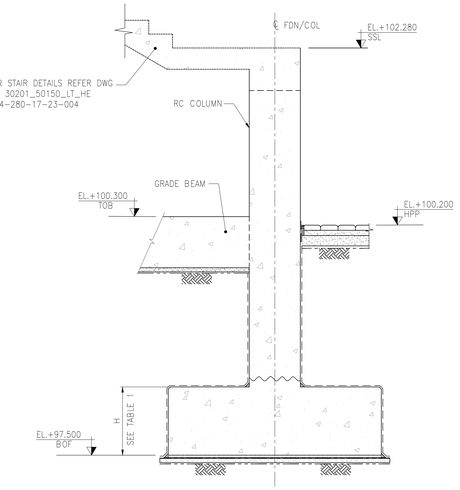


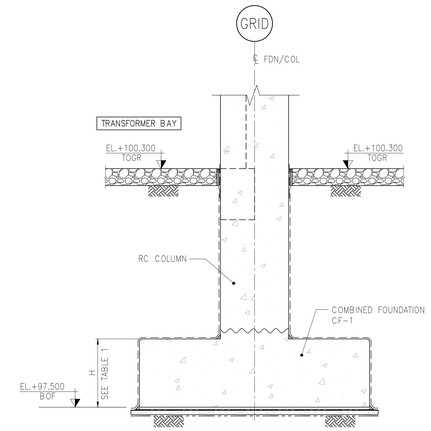
A 02 SECTION (FORMWORK)



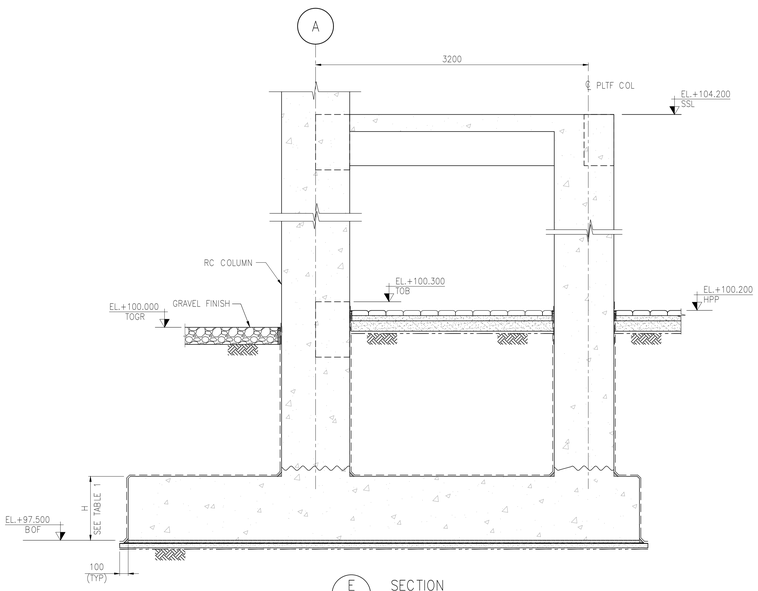
B 02 SECTION (FORMWORK)



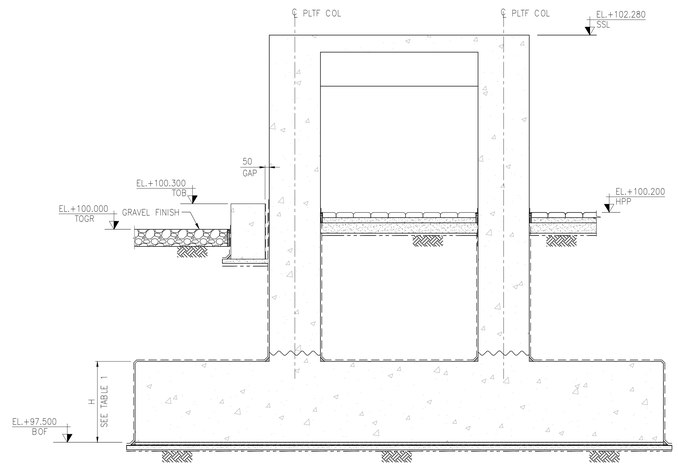
C 02 SECTION (FORMWORK)



D 02 SECTION (FORMWORK)



E 02 SECTION (FORMWORK)



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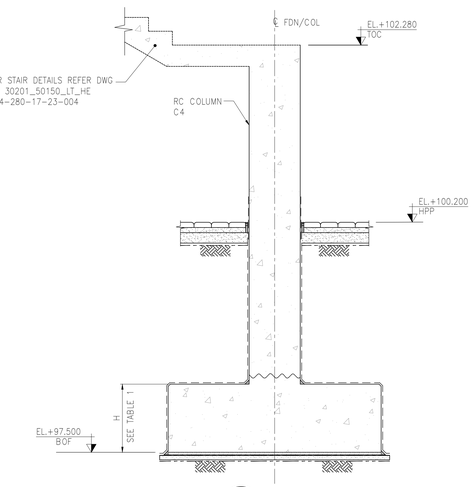

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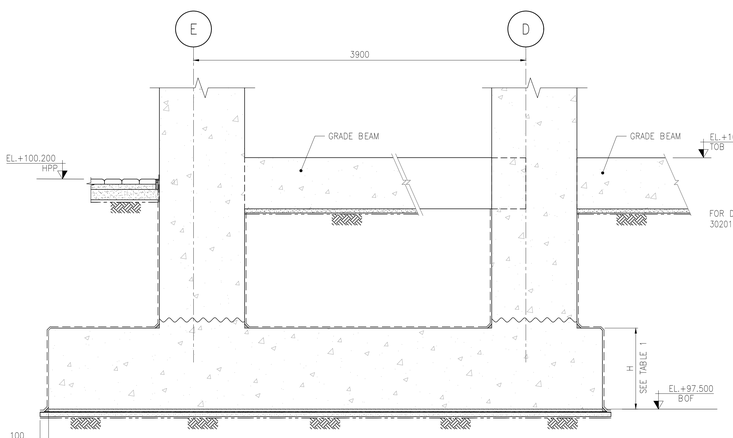

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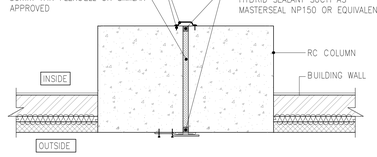
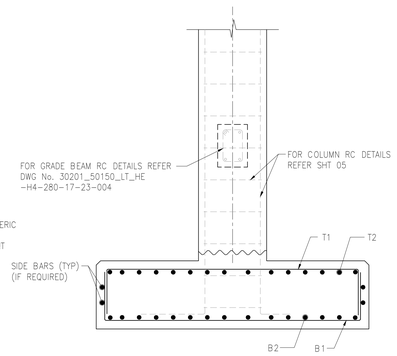


TYP EXPANSION JOINT AT COLUMNS



TYPICAL REINFORCEMENT DETAILS FOR FOOTING TYPES F1–F13

TABLE–1

FOUNDATION MKD	QUANTITY	FOOTING DIMENSIONS		REINFORCEMENT				SIDE BARS	REMARKS
				AT TOP		AT BOTTOM			
		L x B	H	ALONG LENGTH (T1)	ALONG BREADTH (T2)	ALONG LENGTH (B1)	ALONG BREADTH (B2)		
F1	1	6500 x 4000	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F1a	1	6500 x 4000	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F2	5	3200 x 3200	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F2a	1	3400 x 3400	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F3	5	4000 x 4000	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F4	8	3400 x 3400	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F5	1	4400 x 4400	900	T16–150	T16–150	T20–150	T20–150	3–T12EF	
F5a	1	5000 x 5400	900	T16–150	T16–150	T20–150	T20–150	3–T12EF	
F6	3	2500 x 2500	700	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F7	8	4000 x 4000	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F8	2	4000 x 4000	900	T16–150	T16–150	T20–150	T20–150	3–T12EF	
F9	3	4000 x 6000	750	T16–150	T16–150	T20–150	T20–150	2–T12EF	
F10	2	3000 x 6200	900	T16–150	T16–150	T20–150	T20–150	3–T12EF	
F11	1	5000 x 6500	900	T16–150	T16–150	T20–150	T20–150	3–T12EF	
F12	7	3000 x 5500	600	T16–150	T16–150	T16–150	T16–150	–	
F13	4	3000 x 9500	600	T16–150	T16–150	T16–150	T16–150	–	
CF1	1	(SEE LAYOUT)	800	(SEE PLAN)				2–T12EF	

– FOR FOUNDATION ORIENTATION REFER SHT 02

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-003 SHT 02.

HOLD

1. GRADE BEAM ELEVATION

SCALE 1:25 (0 – 0.5 – 1.0 – 1.5 – 2.0 – 2.5m)

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
O	29-JAN-2026	ISSUED FOR CONSTRUCTION (IFC)	VA	KCH/BH	VV/MS
B	17-JAN-2026	ISSUED FOR APPROVAL (IFA)	JAM	KCH/BH	VV/MS
A	23-DEC-2025	ISSUED FOR COMPANY REVIEW (CR)	SUT	KCH/BH	VV/MS

SCALE: 1:25 | DRAWN: SAULE | CHKD: ANIRBAN/BALAJI | APPRD: VISWANATH/SDFI

ADNOC Gas

EPCm CONTRACTOR: wood. | EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5) LONG TERM - EXECUTE STAGE EPCm FOR HABSHAN DBN

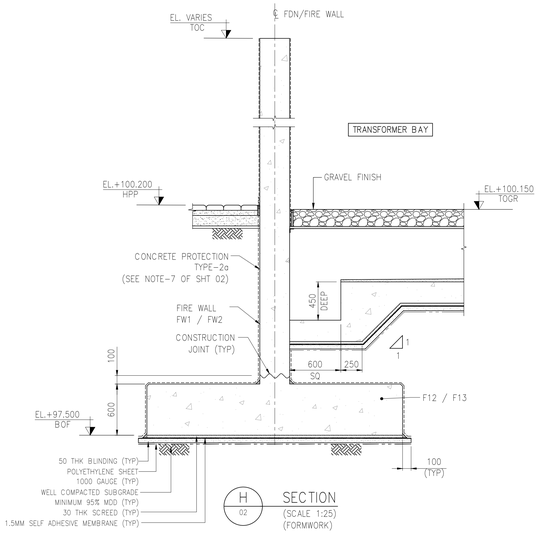
DRAWING TITLE: HABSHAN 4 – UNIT 262 FOUNDATION GA & DETAILS SUBSTATION SS-203 FOUNDATION SECTIONS & DETAILS

UNIT: 262 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No.: 30201_50150_LT_HE | ADNOC GAS DWG. No.: 30201_50150_LT_HE-H4-280-17-23-003 | SHEET: 03 OF 05 | REV: O

SHEET SIZE : A0

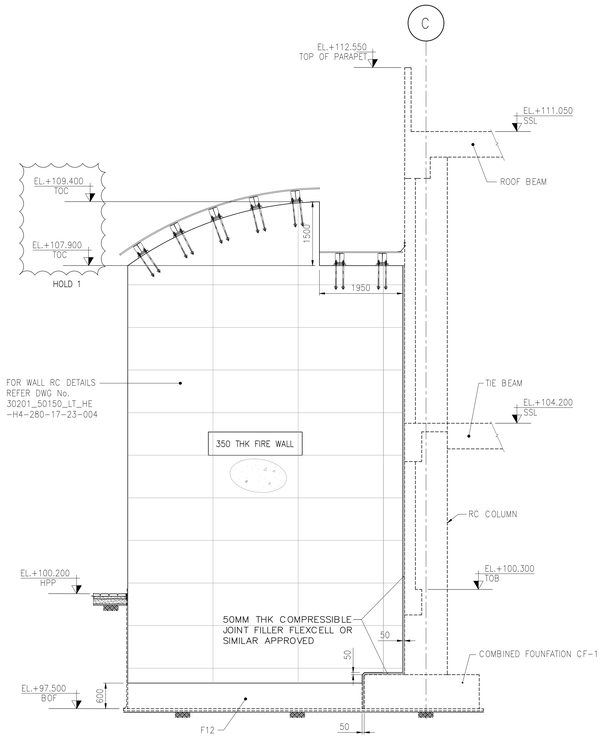
CAD NO. 30201_50150_LT_HE-H4-280-17-23-003_O.dwg

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER.

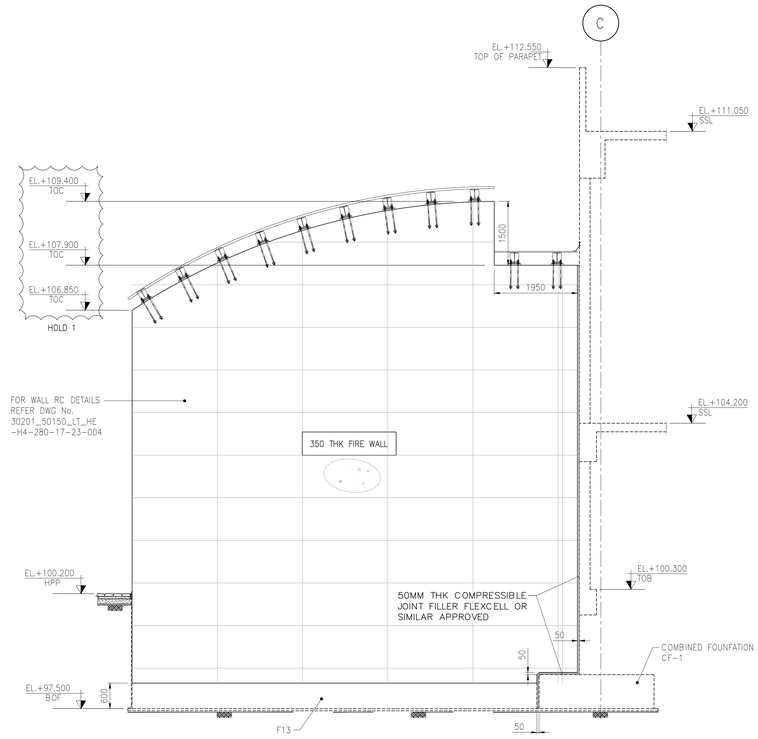


H 02 SECTION
(SCALE 1:25)
(FORMWORK)

TYPICAL FIRE WALL SECTION



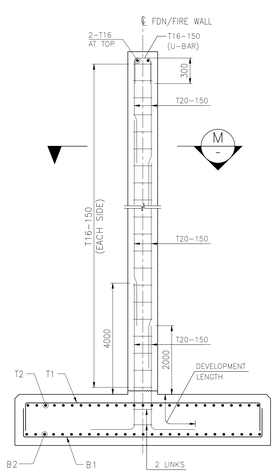
K 02 SECTION
(SCALE 1:50)
(FORMWORK)



L 02 SECTION
(SCALE 1:50)
(FORMWORK)

WALL CONCRETE = 488.00 CUM. (APPROXIMATE)

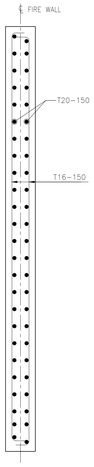
NOTE : ALL WALL QUANTITIES INCLUDED. FOR ALL FOUNDATION QUANTITIES SHALL BE REFEREED TO RESPECTIVE DRAWINGS.



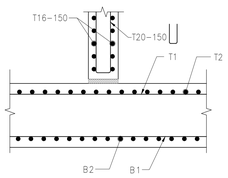
H 02 SECTION
(SCALE 1:25)
(REINFORCEMENT)

TYPICAL FIRE WALL RC DETAIL

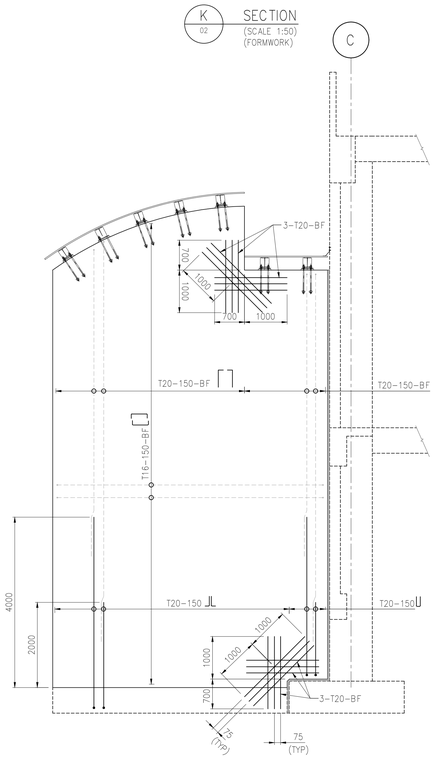
ONLY STARTER REBARS OF FIREWALL CAN BE CONSTRUCTED WITH THIS DRAWING



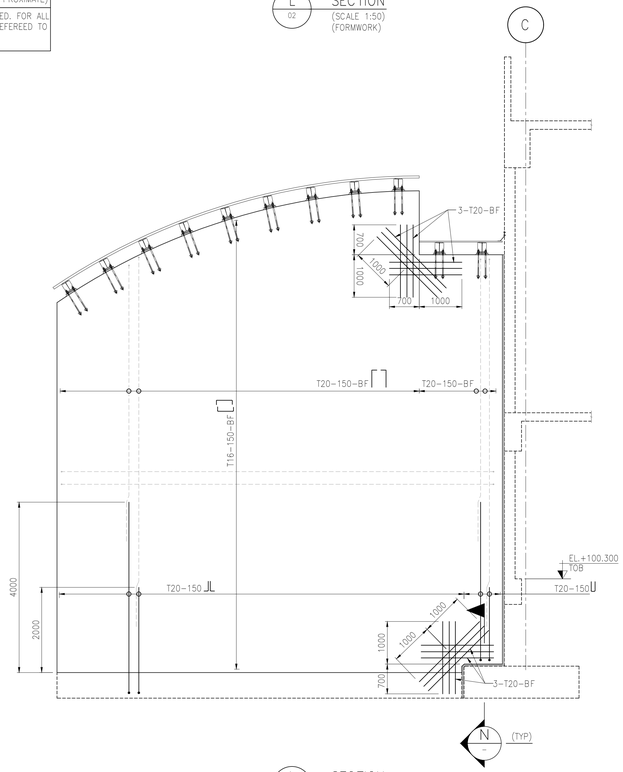
M – SECTION
(SCALE 1:25)
(REINFORCEMENT)



N – SECTION
(SCALE 1:25)
(REINFORCEMENT)



K 02 SECTION
(SCALE 1:50)
(REINFORCEMENT)



L 02 SECTION
(SCALE 1:50)
(REINFORCEMENT)

NOTES

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2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-003 SHT 02.
4. DEVELOPMENT LENGTH LD SHALL BE MINIMUM 50xBAR DIA SPLICE LAPPING LENGTH SHALL BE 1.3 x LD. THE CENTER TO CENTER OF SPLICE STAGGERING SHALL NOT BE LESS THAN 1.3 LD.
5. TYPICAL SPLICE LOCATION PROVIDED BASED ON THAT SUITABLE SPLICE LOCATION SHALL BE ARRIVED DURING BAR BENDING SCHEDULE WHEREVER NOT SHOWN IN THE REINFORCEMENT PLAN. 50% OF REBAR SHALL BE SPLICED AT ONE LOCATION.

HOLD

1. HEIGHT OF FIRE WALL
2. INSERT PLATE LOCATION

SCALE 1:25 — 0 0.5 1.0 1.5 2.0 2.5m

SCALE 1:50 — 0 1.0 2.0 3.0 4.0 5.0m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	29-JAN-2026	ISSUED FOR CONSTRUCTION (IFC)	VA	KCH/BH	VV/MS
B	17-JAN-2026	ISSUED FOR APPROVAL (IFA)	JAM	KCH/BH	VV/MS
A	20-DEC-2025	ISSUED FOR COMPANY REVIEW (CR)	SUT	KCH/BH	VV/MS

SCALE: 1:25 | DRAWN: SAULE | CHKD: ANIRBAN/BALAJI | APPRD: VISWANATH/SOFI

أدنوك للغاز
ADNOC Gas

EPCm CONTRACTOR: wood. AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

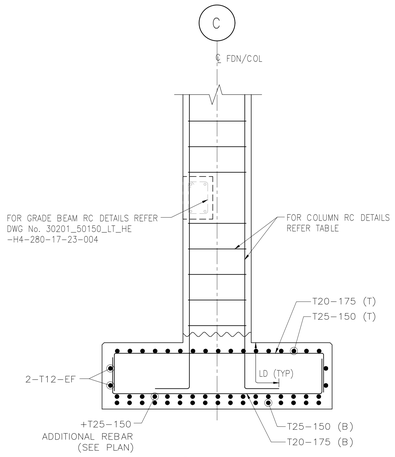
DRAWING TITLE: HABSHAN 4 – UNIT 262
FOUNDATION GA & DETAILS SUBSTATION SS-203
FIRE WALL DETAILS

UNIT: 262 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No. 30201_50150_LT_HE | ADNOC GAS DWG. No. 30201_50150_LT_HE-H4-280-17-23-003 | SHEET 04 OF 05 | REV 0

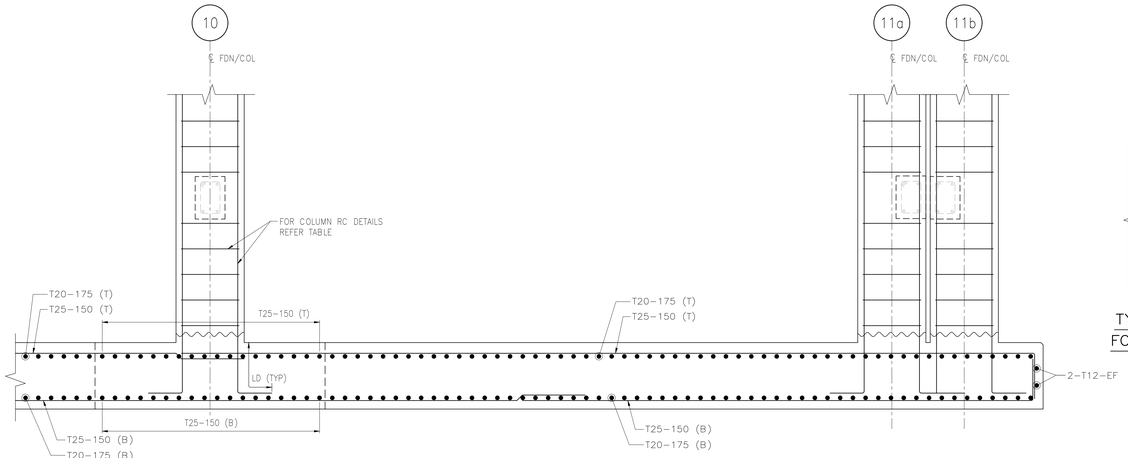
SHEET SIZE : A0

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER

CAD NO: 30201_50150_LT_HE-H4-280-17-23-003_C.dwg



P SECTION (FORMWORK)



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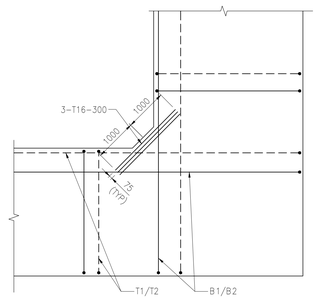
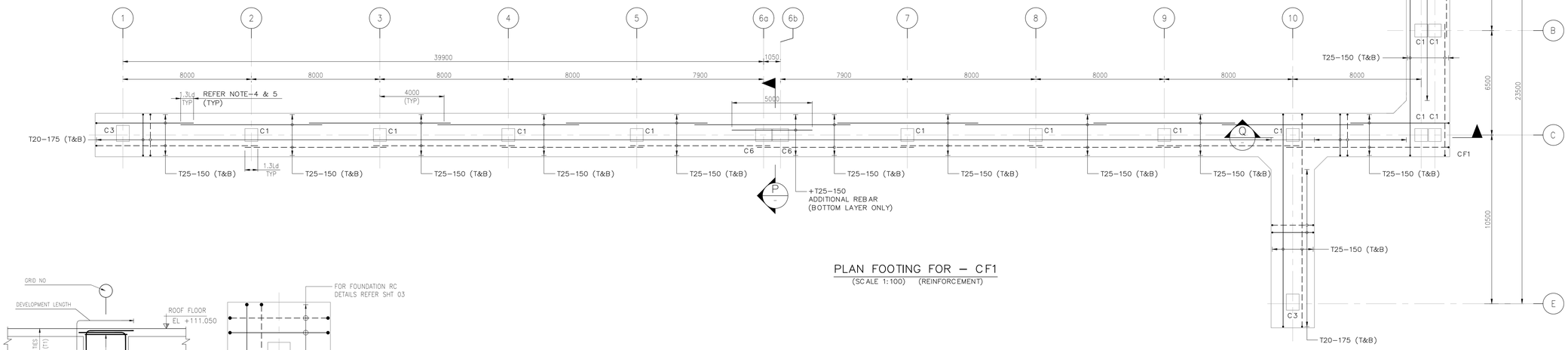


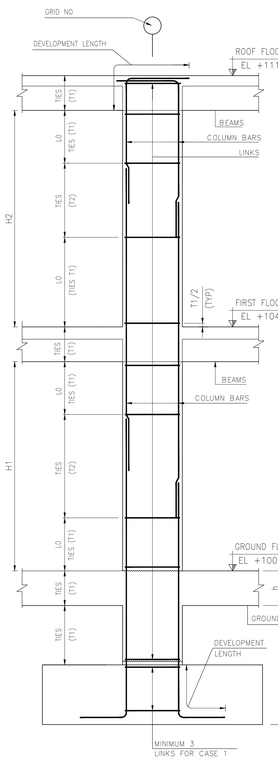
TYPICAL ADD'L REINFORCEMENT DETAIL FOR CHAMFERED LOCATION OF FOOTING

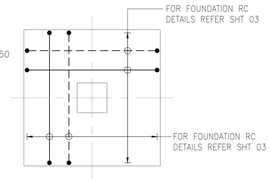
(SCALE: NTS)



PLAN FOOTING FOR – CF1

(SCALE 1:100) (REINFORCEMENT)





TYPICAL PLAN FOOTING FOR – F1 TO F13

(SCALE 1:100) (REINFORCEMENT)

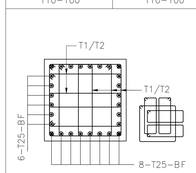
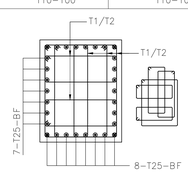
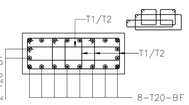
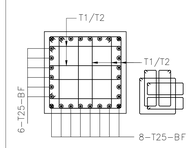
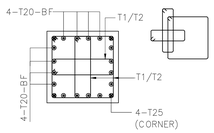
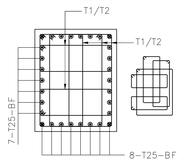
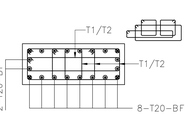
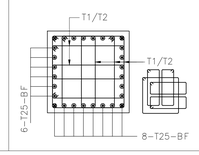
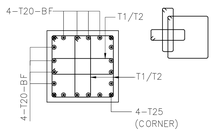
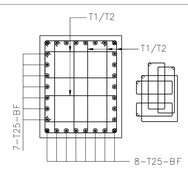
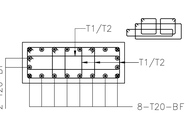
CASE 1: FOR INTERMEDIATE MOMENT FRAMES / SEISMIC DESIGN CATEGORY C

T1 ≤
- a) FOR GRADE 420, MIN OF (8 TIMES SMALLEST LONGITUDINAL REINFORCEMENT BAR DIAMETER, 200mm) FOR GRADE 550, MIN OF (6 TIMES SMALLEST LONGITUDINAL REINFORCEMENT BAR DIAMETER, 150mm)
- b) 0.5 SMALLEST CROSS-SECTIONAL DIMENSION OF THE COLUMN.

T2 – AS PER DESIGN
AT COLUMN SPLICE LOCATION, T2 = MIN (d/4, 100mm), WHERE d IS COLUMN LEAST DIMENSION

LO : GREATER OF THE FOLLOWING
- a) MAX COLUMN DIMENSION
- b) 1/6 OF CLEAR HEIGHT OF COLUMN
- c) 450mm

COLUMN REINFORCEMENT SCHEDULE

FLOOR \ COLUMN ID	C1 (800x800)		C2 (700x700)		C3 (800X1000)		C4 & C4a (500X500)		C5 (1000X400)		C6 (1000X800)	
	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2
	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100
FIRST FLOOR BEAM TO ROOF BEAM (EL +104.200 TO +111.050)												
GRADE BEAM TO FIRST FLOOR BEAM (EL +100.300 TO +102.280 FOR COLUMN C4) (EL +100.300 TO +104.200 FOR COLUMNS C2, C4a & C6)												
FOUNDATION TO GRADE BEAM (EL +97.500 TO +100.300)												

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3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-003 SHT 02.
4. DEVELOPMENT LENGTH LD SHALL BE MINIMUM 50xBAR DIA SPLICE LAPPING LENGTH SHALL BE 1.3 x LD. THE CENTER TO CENTER OF SPLICE STAGGERING SHALL NOT BE LESS THAN 1.3 LD.
5. TYPICAL SPLICE LOCATION PROVIDED BASED ON THAT SUITABLE SPLICE LOCATION SHALL BE ARRIVED DURING BAR BENDING SCHEDULE WHEREVER NOT SHOWN IN THE REINFORCEMENT PLAN. 50% OF REBAR SHALL BE SPLICED AT ONE LOCATION.

LEGEND

- TOP BAR
- BOTTOM BAR

SCALE 1:25 (0, 0.5, 1.0, 1.5, 2.0, 2.5m)

REV.	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	29-JAN-2026	ISSUED FOR CONSTRUCTION (IFC)	VA	KCH/BH	VV/MS
B	17-JAN-2026	ISSUED FOR APPROVAL (IFA)	JAM	KCH/BH	VV/MS
A	05-DEC-2025	ISSUED FOR COMPANY REVIEW (CR)	SUT	KCH/BH	VV/MS

SCALE: 1:25 | DRAWN: SAULE | CHKD: ANIRBAN/BALAJI | APPRD: VISWANATH/SOFI

ADNOC Gas — أدنوك للغاز — ADNOC

EPCm CONTRACTOR: wood. — EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5) LONG TERM - EXECUTE STAGE EPCm FOR HABSHAN DBN

DRAWING TITLE: HABSHAN 4 – UNIT 262 FOUNDATION GA & DETAILS SUBSTATION SS-203 COLUMN DETAILS

UNIT: 262	PLANT: H4
CONTR. PROJ. No. 3000076501	CONT. DWG. No.: XXXXXXX
ADNOC GAS CONTRACT No. 3000076501	
ADNOC GAS PROJ. No. 30201_50150_LT_HE	ADNOC GAS DWG. No. 30201_50150_LT_HE-H4-280-17-23-003
SHEET 05 OF 05	REV. 0

SHEET SIZE : A0

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CAD NO: 30201_50150_LT_HE-H4-280-17-23-003_C.dwg